

Content Disarm and Reconstruction as an Adversarial Defense for Vision-Language Models: An Empirical Audit of Deployed CDR Tools

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1 Project Introduction

1.1 Background / Scope

Vision-Language Models (VLMs) such as LLaVA [1] couple a CLIP-style Vision Transformer [2] to a language model. This visual channel is vulnerable to *pixel-space adversarial perturbations*: a small, bounded change to the image, optimized end-to-end through the model, can steer the generated text [3, 4]. A common, model-agnostic defense family is *input transformation*: re-encoding or reconstructing the image before it reaches the encoder [5].

Independently, the security industry deploys *Content Disarm and Reconstruction (CDR)*: parse a file, discard non-conforming or active structure (metadata, embedded objects, anomalous chunks), and rebuild a clean file from validated content. Real, reproducible open-source CDR tools already exist. *ICDR* [6, 7] has a published pipeline of **transCode** (re-encode, strip all metadata) $\rightarrow$ **resize** (97% compress/restore) $\rightarrow$ **AdvFilter** (Gaussian blur then sharpen). *Dangerzone* [8] renders a document to raw RGB pixels and rebuilds a clean file, discarding *all* non-pixel structure. This project evaluates these existing implementations.

Scope. This is a computer-vision robustness study: we audit whether deployed image-CDR tools, applied as a pre-encoder defense, reduce the success of targeted adversarial images against a VLM while preserving legitimate visual content, and whether they do better than trivial JPEG re-encoding. The methods are core CV: adversarial optimization through a ViT-based encoder, frequency-domain analysis of what reconstruction removes, image-fidelity measurement, and vision-encoder robustness evaluation.

1.2 Problem Statement / Motivation

A named security framework, SAF-M-49 [9], *recommends* multimedia CDR as a mitigation against multimodal prompt injection, but it provides no empirical validation (it lists efficacy and latency measurement as future work). Meanwhile, existing CDR-for-AI research sanitizes *model weight files* [10, 11], not input images, and image-CDR research targets *malware / steganography removal* [7], not adversarial robustness. Purpose-built adversarial *purification* for VLMs does exist, but it is bespoke machine learning (diffusion or latent-space purifiers) [12–14], not the deterministic security tools practitioners actually deploy. This leaves a concrete, testable question: *does a deployed CDR tool, recommended for exactly this purpose, actually blunt adversarial images on a VLM, and is it worth more than a one-line JPEG call?* My professional background in CDR motivates a faithful setup and a mechanistic reading of *which* disarming operation carries any effect.

2 Related Work / Literature Survey

We separate seven categories to avoid conflation. **(1) CDR on model files/weights** [10, 11]: different surface. **(2) CDR on input images** [6–8]: real tools, but evaluated only for malware/steganography, never adversarial robustness. **(3) Conventional transform defenses** [5, 15]: single-operation (mostly JPEG), classifier-era. **(4) Purification defenses for VLMs** [12–

14]: numerous but bespoke ML, not deployed security tools. (5) **Adversarial attacks on VLMs** [3, 4]: supply our attack methodology. (6) **Typographic / prompt injection** [16, 17]: out of MVP scope (see §5). (7) **Robust-evaluation methodology** [18–20]: governs our adaptive-attack and statistics choices.

Positioned gap (narrow, supported). We are not aware of any empirical evaluation of *deployed image-CDR tools* (ICDR, Dangerzone) as a pre-encoder adversarial defense for VLMs, nor of a test of whether they provide robustness *beyond plain JPEG*. We make no “first/novel” claim about CDR as a concept: SAF-M-49 already recommends it, and the contribution is the missing audit.

3 Project Content Outline

3.1 Core Idea

Insert a deployed CDR tool as a pre-encoder stage, $\text{image} \rightarrow \text{CDR} \rightarrow \text{VLM}$, and measure targeted attack success and clean-task utility against two references: no defense, and a JPEG quality sweep that traces the security–utility frontier. Then ask whether CDR lies beyond that frontier, and *why*.

3.2 Research Questions

- **RQ1.** Does a deployed CDR tool reduce the targeted attack success rate of adversarial images on a VLM while preserving clean VQA accuracy?
- **RQ2.** Does CDR provide robustness *beyond* the JPEG quality-sweep frontier at comparable utility cost?
- **RQ3 (mechanism).** Is any robustness explained by removal of high-frequency perturbation energy, i.e. does the causal chain $\text{CDR operation} \rightarrow \text{perturbation-energy removal} \rightarrow \text{attack failure}$ hold?
- **RQ4 (extension).** Can a CDR-aware adaptive attacker recover attack success, indicating the non-adaptive result overstates robustness?

3.3 Threat Model

- **Attacker.** White-box on the target VLM (following Qi et al. [3]); may perturb the input image only, under an L_∞ budget ε ; optimizes teacher-forcing cross-entropy from raw pixels to LM logits toward a predetermined target answer.
- **Knowledge (two settings).** *Defense-unaware* (MVP: attack then sanitize) and *defense-aware* (Extension: attack *through* the CDR pipeline).
- **Success criterion (single, primary).** *Targeted VQA-answer flip*: for a controlled single-object question with known ground truth, the model outputs the attacker’s predetermined incorrect label. No harmful-content generation.
- **Defender.** Applies one sanitizer before the encoder; compared with no defense and with JPEG.

3.4 Dataset

200 images from the ImageNet-1k validation set [21]: 20 classes $\times$ 10 images (fixed seed list, IDs logged). Each image gets one controlled prompt (“What is the main object? Answer with a single label.”) with the ImageNet class as ground truth and one *predetermined* target label (fixed class-to-target map). Images are resized to the model’s native input (336 px for LLaVA). The *same* images pass through *every* defense (paired design). Attacks are optimized per image on the test image itself, which is standard for per-example adversarial attacks. Because no model is trained, there is no train/test leakage, only per-image white-box optimization. Clean-utility is measured on the unattacked images under each defense.

3.5 Models

Primary: LLaVA-1.5-7B (the Qi-et-al. lineage, maximizing comparability and reproducibility). No CLIP surrogate is used; the attack runs end-to-end on the target model. **Compute budget:** a single 24 GB GPU; white-box PGD through 7B parameters uses fp16 and gradient checkpointing. **Fallback** for < 16 GB: Qwen2-VL-2B [22] as primary. A second model is used *only* in the extension to test generalization, not in the MVP.

3.6 Experimental Design

Defenses (minimal, principled): D0 none; D1--D3 JPEG at quality {90, 75, 50} (traces the frontier); D4 ICDR (`final_fun`: `transCode`→`resize`→`AdvFilter`) [6]; D5 Dangerzone (full render-and-rebuild) [8]. **Ablation (conditional).** *Only if* ICDR (D4) differs significantly from the JPEG frontier do we decompose its published operations (`transCode` / `resize` / `AdvFilter` / the alternate LSB-clean and gamma steps) to localize the effect. This keeps the matrix small unless the data justify expansion. **Comparison protocol.** Plot every defense in (clean-accuracy-retained, robust-accuracy-under-attack) space; JPEG {90, 75, 50} traces the frontier; CDR tools appear as points. RQ2 asks whether CDR lies *on* or *beyond* that frontier. Fidelity metrics are reported as descriptive covariates, not as a matching axis.

3.7 Step-by-Step To-Do Plan

1. Load LLaVA-1.5-7B; wrap it as a differentiable pixel→logits function.
2. Integrate deployed defenses: JPEG (Pillow), ICDR [6], Dangerzone [8]; verify each runs deterministically (log versions/configs).
3. Build the 200-image ImageNet VQA benchmark + fixed target-label map.
4. Implement the targeted PGD attack (teacher forcing to target answer), $\varepsilon \in \{4, 8, 16\}/255$; validate attack strength on D0.
5. Run the paired ladder: attack $\times$ {D0--D5} on the primary model. ← **MVP result.**
6. Mechanistic analysis (RQ3): residual perturbation energy and its frequency spectrum before/after each defense.
7. Statistics: McNemar per defense pair, bootstrap CIs, Holm correction.
8. **Extension:** defense-aware adaptive attack (§3.6, §6); optional second model for generalization.
9. Aggregate, plot the frontier, build the demo, write up.

4 Results

4.1 Expected Outcomes

A security–utility frontier that places deployed CDR tools relative to JPEG and to no defense, on one VLM with a targeted white-box attack; a mechanistic test of *why* any effect occurs; an optional adaptive-attack check; and a reproducible codebase and demo.

4.2 Evaluation Metrics (each tied to one RQ)

Metric	Research question it answers
Targeted ASR ($\pm 95\%$ bootstrap CI)	RQ1, RQ2, RQ4: does the defense stop the attack?
Clean VQA accuracy under each defense	RQ1: does the defense preserve utility?
Security–utility frontier plot	RQ2: does CDR beat JPEG at equal utility cost?
Residual perturbation energy + FFT spectrum	RQ3: does robustness come from removing high-freq perturbation?
PSNR / SSIM / LPIPS (descriptive)	context for the frontier (not a matching axis)
CDR/JPEG latency	practical cost of deployment

4.3 Experimental Evaluation

Paired design (identical images through all defenses) $\Rightarrow$ **McNemar’s test** for each pairwise ASR comparison; **95% bootstrap CIs** on ASR and clean accuracy; **Holm–Bonferroni** across the small set of comparisons; effect size reported as the ASR difference. Attack validity is checked per Carlini et al. [20] (attack must reach high ASR on D0 before any defense claim).

4.4 Visual Demo

Panels of (clean $\rightarrow$ correct answer), (adversarial $\rightarrow$ target answer), (adversarial $\rightarrow$ CDR $\rightarrow$ recovered or not), each annotated with its FFT spectrum and the residual-energy readout; plus the frontier plot with the CDR points marked.

4.5 Hypotheses and Falsification

- **H1:** CDR reduces targeted ASR vs. D0. *Falsified if* ASR under D4/D5 is within the CI of D0.
- **H2:** CDR lies beyond the JPEG frontier for ≥ 1 operating point. *Falsified if* CDR points sit on/inside the JPEG curve.
- **H3:** Robustness tracks high-frequency perturbation removal. *Falsified if* defenses that remove comparable high-freq energy give divergent ASR, or vice versa.
- **H4:** Aggressive CDR (Dangerzone) costs measurable clean accuracy. *Falsified if* clean accuracy under D5 is within D0’s CI.
- **H5:** A defense-aware attack recovers substantial ASR. *Falsified if* adaptive ASR stays near non-adaptive defended ASR.

5 Scope and Limitations

This project is an audit, not a new defense method. The MVP studies *one* attack family (pixel-space perturbation), *one* primary VLM, and *deployed* CDR tools. Commercial CDR (e.g. MetaDefender, Votiro) is proprietary and not bit-reproducible, so it is out of scope. Success is a targeted-VQA proxy for output manipulation, not harmful-content generation. Attacks are per-image white-box; results bound the white-box case, the strongest for the defender to survive. On the available 16 GB GPU the model is loaded in 4-bit (NF4) rather than fp16, since the fp16 weights leave no room for the input-gradient backward pass; the attack is still white-box, end-to-end through the quantized model. A fp16 run on a ≥ 24 GB GPU is a stretch goal. Typographic and visual prompt-injection is deliberately excluded from the MVP and left as an optional contrast experiment. Rendered text is legitimate visible content, so CDR is expected *not* to remove it, and Dangerzone’s OCR layer could even re-inject it. We would only spot-check this, not build the project on it.

6 Risks and Fallbacks

Risk	Fallback
White-box PGD through 7B exceeds GPU memory	Gradient checkpointing + fp16; else Qwen2-VL-2B as primary.
ICDR/Dangerzone hard to script in the pipeline	Run them as external processes on saved images (batch), then load; both are open-source and file-based.
CDR indistinguishable from JPEG	That <i>is</i> the RQ2 answer (H2 falsified), a useful negative result for practitioners.
Adaptive attack (BPDA) too heavy through 7B + CDR	Extension only; simpler fallback is a defense-aware transfer attack through a differentiable CDR approximation (differentiable-JPEG + straight-through for non-differentiable steps).
Attack too weak on D0	Increase ε /iterations until high ASR on D0 before any defense claim [20].
Time runs out	MVP (RQ1+RQ2, one model, one attack, 200 images) stands alone.

7 Timeline

Week	Deliverable
W1	Differentiable VLM wrapper; integrate JPEG/ICDR/Dangerzone; benchmark built.
W2	Targeted PGD implemented and validated on D0; paired ladder run. ← MVP result.
W3	Mechanistic analysis (residual energy, FFT) + statistics (McNemar, CIs, Holm) + frontier plot.
W4	<i>Extension (optional)</i> : adaptive attack and/or second-model generalization; demo; write-up.

Appendix A: Conservative Novelty Statement

Image Content Disarm and Reconstruction is an existing, deployed file-sanitization technique, with open-source implementations such as ICDR [6, 7] and Dangerzone [8], and security guidance already recommends multimedia CDR against multimodal prompt injection [9]. However, that recommendation is unvalidated, existing CDR-for-AI research sanitizes model weight files rather than model inputs [10, 11], and purpose-built adversarial purification for VLMs is bespoke machine learning rather than a deployed security tool [12, 13]. We therefore do not claim CDR is a novel idea. Our contribution is an empirical audit: we are not aware of prior work that evaluates deployed image-CDR tools as a pre-encoder adversarial defense for vision-language models, or that tests whether they provide robustness beyond plain JPEG re-encoding, with a controlled targeted white-box attack, a security–utility frontier comparison, a mechanistic frequency-domain analysis, and an adaptive-attack robustness check.

Appendix B: Experimental Matrix

Attacks / Model	Primary metrics
D0 none; D1--D3 JPEG{90, 75, 50}; D4 ICDR; D5 Dangerzone. Model: LLaVA-1.5-7B. space tar- geted PGD $(L_\infty,$ $\varepsilon \in$ $\{4, 8, 16\}/255),$ white- box on the VLM [3]	Targeted ASR $(\pm CI);$ clean VQA ac- cu- racy; resid- ual per- tur- ba- tion en- ergy + FFT; PSNR/SSIM/LP la- tency
D4/D5; LLaVA-1.5-7B (+ Qwen2-VL-2B for generalization) (defense- aware) <i>[Ex- ten- sion]</i>	Adaptive ASR vs. non- adaptive de- fended ASR

Appendix C: MVP vs. Extension

MVP (must implement; a complete result on its own): differentiable LLaVA-1.5-7B wrapper; integrated deployed defenses (JPEG sweep, ICDR, Dangerzone); 200-image ImageNet VQA benchmark; targeted white-box PGD validated on D0; paired ladder D0--D5; security-utility frontier; mechanistic residual-energy/FFT analysis; paired statistics; tables, frontier plot, and demo. Answers RQ1–RQ3 and remains interesting even if CDR = JPEG.

Extension (optional): defense-aware adaptive attack [18] testing for a false sense of security (RQ4); generalization to a second VLM; a small typographic-injection contrast check.

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